CREDINTEL

CREDIT INTELLIGENCE & RISK ANALYTICS

PROJECT REPORT

Transforming Loan Data into Actionable Lending Insights

4,269	62.22%	5
Loan applications	Approval rate	Dashboard pages

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Python • Pandas • NumPy • MySQL • Power BI

1. Executive Summary

CredIntel is an end-to-end credit risk analytics project examining loan application outcomes and borrower financial profiles. It uses 4,269 loan application records and combines Python-based data exploration, financial feature engineering, MySQL business analysis, and a five-page Power BI dashboard.

The dataset contains 2,656 approved applications (62.22%) and 1,613 rejected applications (37.78%). Average CIBIL score is 703.46 for approved applications and 429.47 for rejected applications. These are descriptive results from the supplied dataset; they indicate association, not causation.

Project at a glance

Dimension	Details
Domain	Credit risk, lending analytics, borrower financial analysis
Dataset	4,269 records and 13 attributes
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, MySQL, Power BI
Deliverables	Python notebooks, SQL analysis, Power BI report and dashboard screenshots
Analytical approach	Descriptive analytics; no machine-learning model

2. Business Context & Problem Statement

Lenders evaluate applicants with different credit histories, income levels, loan requirements, employment characteristics, and asset positions. Analyzing these dimensions helps describe portfolio composition and observed lending outcomes.

CredIntel treats this as an analytics problem: examine historical applications, compare approved and rejected groups, construct interpretable financial indicators, and communicate results through SQL analysis and interactive reporting. It does not automate lending decisions or build an approval prediction model.

3. Project Objectives

• Measure the distribution of approved and rejected applications.

• Explore CIBIL score patterns across loan outcomes and score segments.

• Analyze income, loan amounts, education, employment, and asset values.

• Create descriptive financial ratios and borrower segments.

• Use MySQL to answer structured lending and portfolio questions.

• Build Power BI dashboards to communicate findings.

4. Dataset Overview

The supplied CSV contains 4,269 records and 13 attributes covering applicant characteristics, annual income, loan amount and term, CIBIL score, asset values, and loan decision status.

Attribute	Description
Applicant income	Annual income of applicant
Loan amount / term	Requested amount and repayment duration
CIBIL score	Applicant credit score
Education / self-employment	Applicant profile categories
Residential / commercial assets	Value of respective asset classes
Luxury / bank assets	Value of respective asset classes
Loan status	Approved or rejected outcome

5. Data Understanding & Quality Assessment

Initial checks covered dataset dimensions, data types, descriptive statistics, missing values, and duplicate rows. The supplied CSV contains 0 missing values and 0 fully duplicated rows according to direct pandas checks.

Data preparation considerations

• Trim leading and trailing whitespace from categorical text before grouping or filtering.

• The dataset contains negative residential asset values; treat these as potential anomalies for investigation rather than automatically deleting them.

• Keep unusual recorded values distinct from missing values.

• Apply CIBIL score bands consistently, including the minimum observed score and boundary values.

6. Exploratory Data Analysis (EDA)

EDA examines the loan outcome distribution and compares borrower characteristics across approval categories. The analytical focus includes CIBIL score, income, loan amount, education, employment, assets, and risk segments.

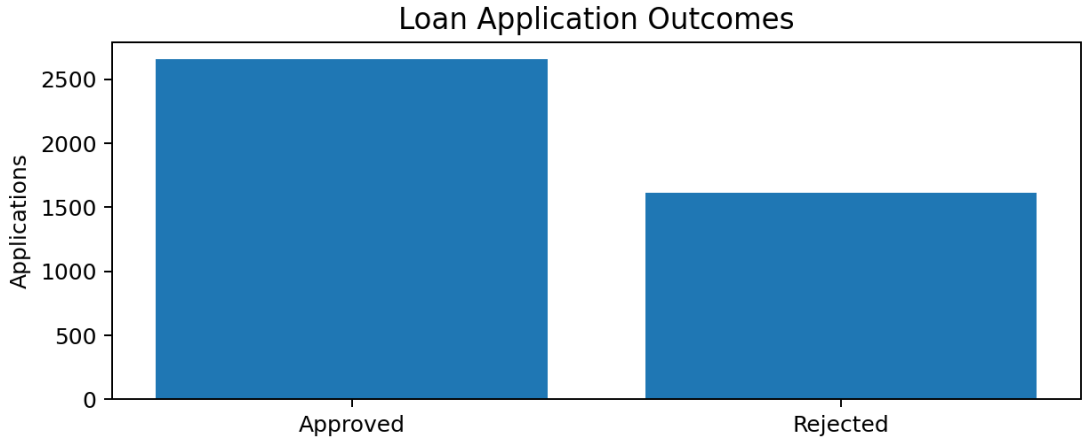


Figure 1. Approved and rejected application counts calculated from the supplied CSV.

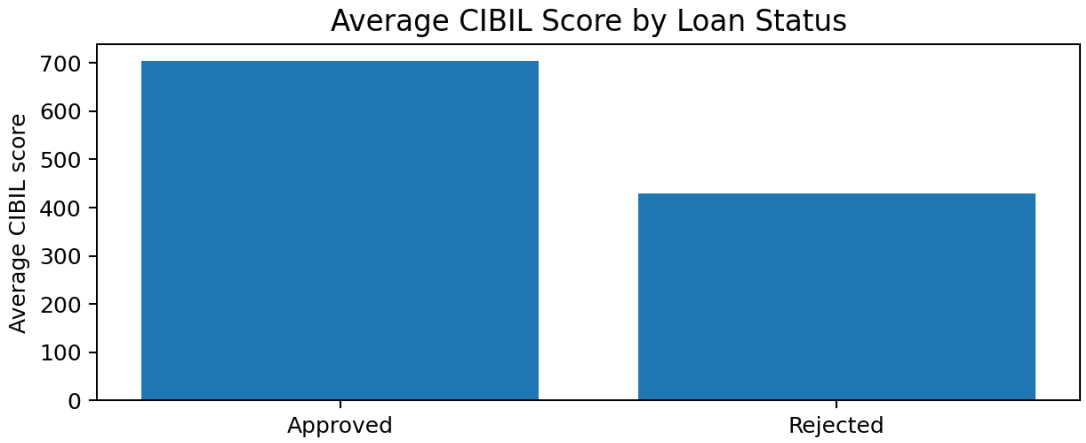


Figure 2. Average CIBIL score by loan status calculated from the supplied CSV.

7. Feature Engineering & Financial Analysis

Derived variables make borrower financial profiles easier to compare. These are descriptive analytical measures, not official lending-policy thresholds.

Feature	Analytical interpretation
Total Assets	Residential + commercial + luxury + bank asset values
Loan-to-Income Ratio	Loan amount divided by annual income
Asset-to-Loan Ratio	Total assets divided by loan amount
Risk Category	Descriptive CIBIL-based borrower grouping
Financial Strength	Grouping based on borrower asset profile
Borrowing Capacity	Exploratory grouping based on loan-to-income ratio

Interpretation notes

• Check ratios for zero denominators and unusual input values.

• Use consistent score-band boundaries when calculating risk categories.

• Quantile-based asset groups describe relative position within this dataset, not universal ratings.

• Borrowing-capacity labels are exploratory and should not be presented as formal credit policy.

8. MySQL Business Analysis

MySQL queries support repeatable business analysis using filtering, grouping, aggregation, conditional expressions, and joins where applicable. The SQL work covers portfolio totals, approval outcomes, average CIBIL score by loan status, applicant categories, and risk segmentation.

Business questions addressed

• What is the overall approval and rejection mix?

• How does average CIBIL score differ by loan status?

• How are applications distributed across borrower categories?

• How do CIBIL-based risk groups relate to observed outcomes?

• What portfolio summaries can support management reporting?

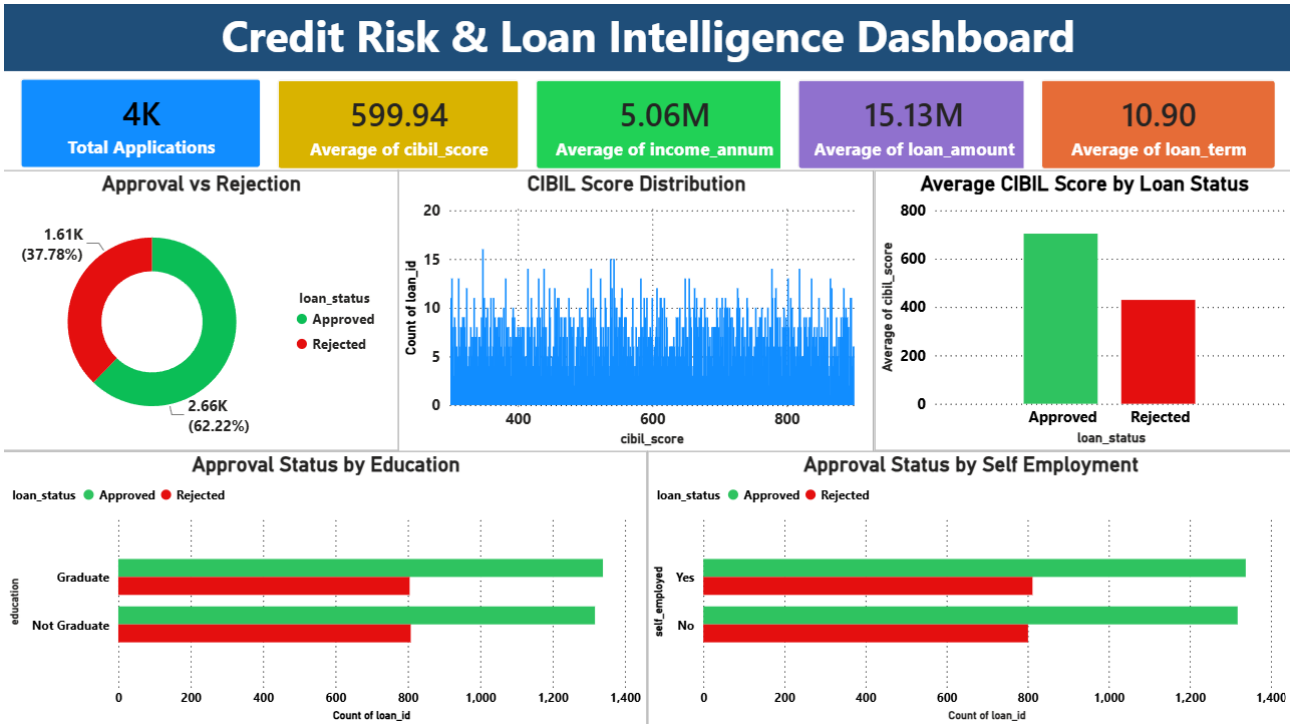
Consistent category cleaning and risk-band definitions are important when comparing results across Python, SQL, and Power BI.

9. Power BI Dashboard

The Power BI report contains five pages that provide complementary views of the loan portfolio. The supplied dashboard screenshots are included in the following pages.

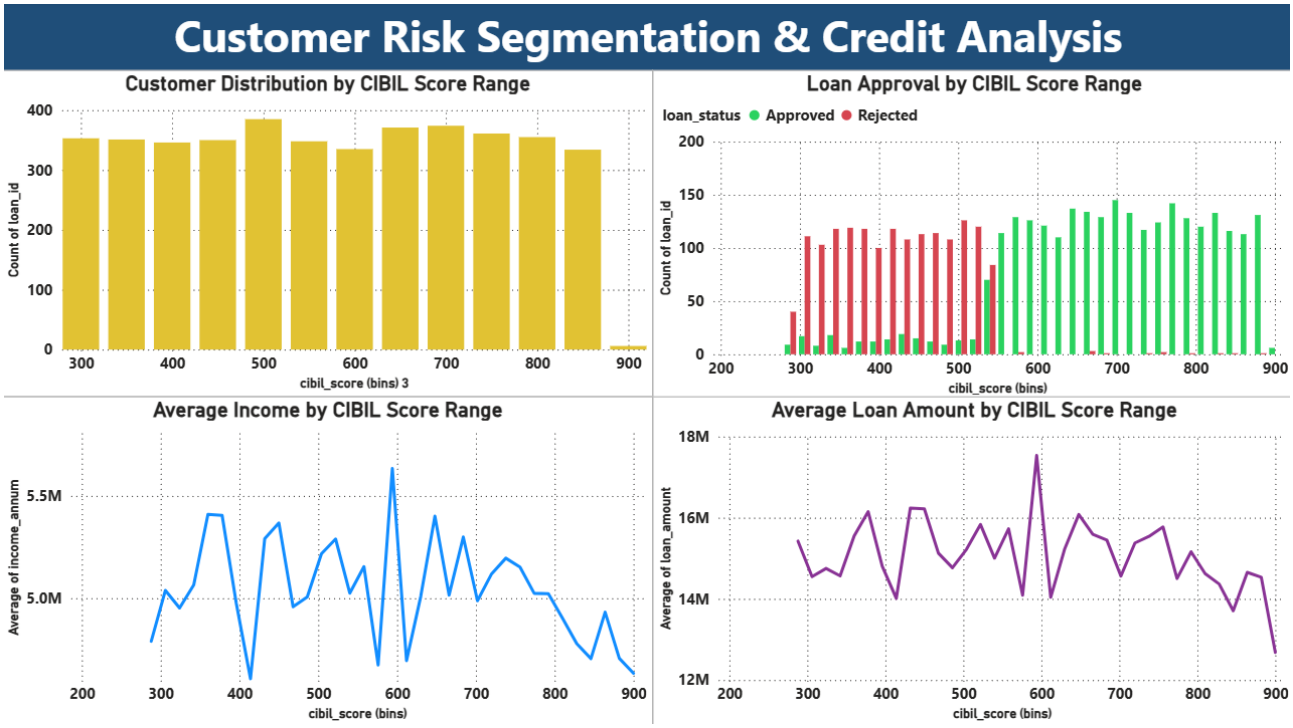
9.1 Credit Risk & Loan Intelligence Dashboard

Portfolio overview: application volume, average CIBIL score, income and loan indicators, approval mix, CIBIL distribution, and applicant category comparisons.



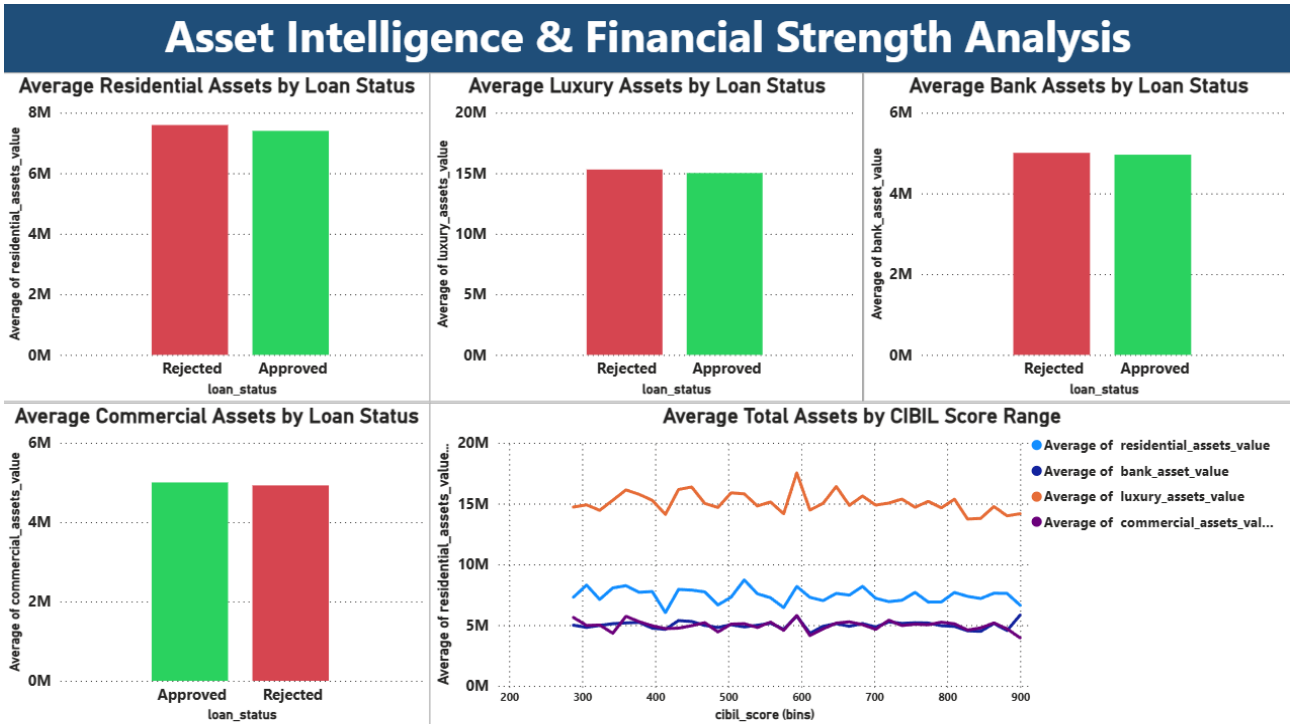
9.2 Customer Risk Segmentation & Credit Analysis

Borrower segmentation and credit analysis, with score-based and applicant-profile comparisons.



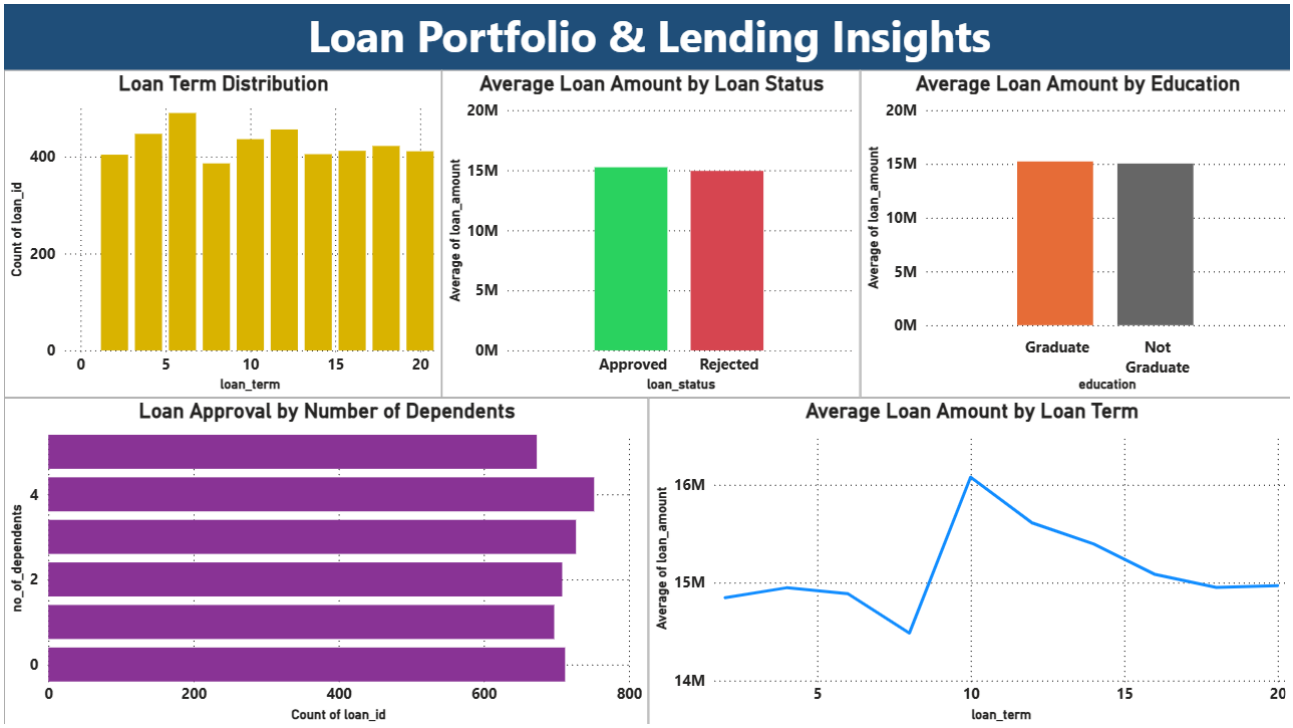
9.3 Asset Intelligence & Financial Strength Analysis

Asset-focused analysis of residential, commercial, luxury, and bank asset dimensions.



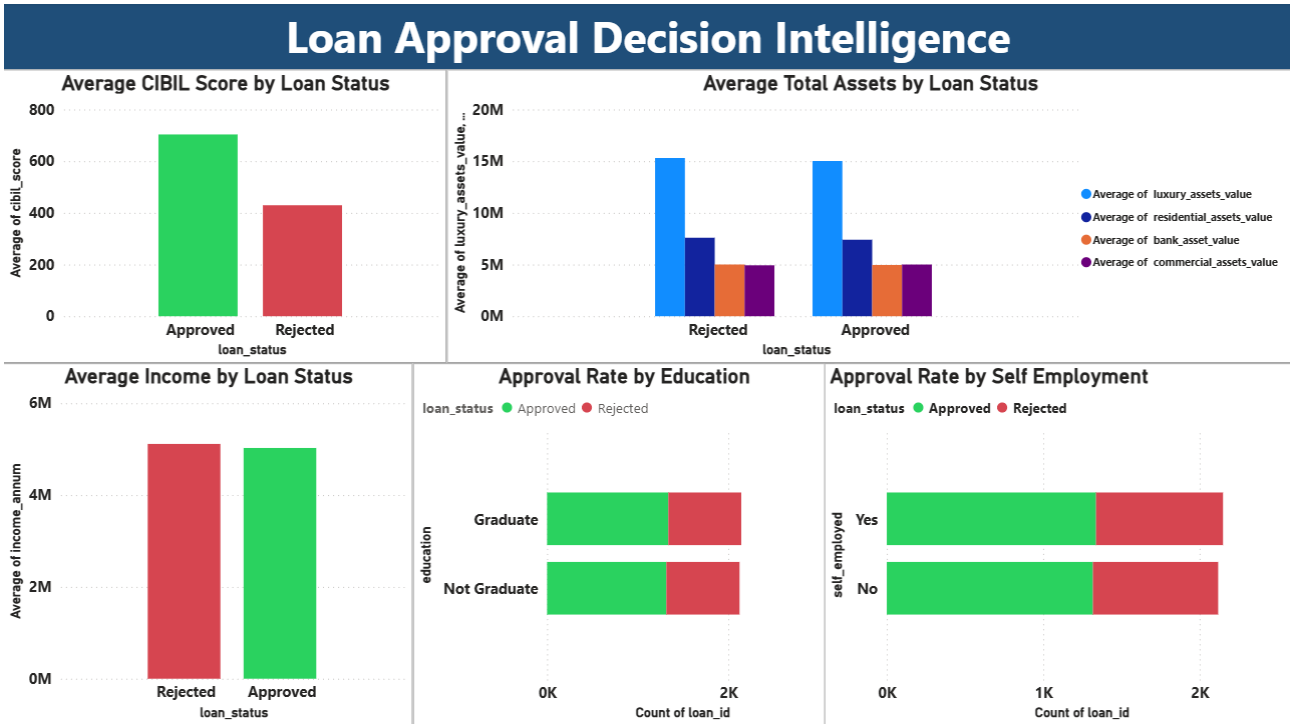
9.4 Loan Portfolio & Lending Insights

Portfolio and lending views bringing together loan characteristics and applicant-level comparisons.



9.5 Loan Approval Decision Intelligence

Approval decision comparisons across applicant characteristics, including education and self-employment.



10. Key Findings & Business Insights

Metric	Observed result
Total applications	4,269
Approved applications	2,656
Rejected applications	1,613
Approval rate	62.22%
Rejection rate	37.78%
Average CIBIL — approved	703.46
Average CIBIL — rejected	429.47
Approval rate for CIBIL > 700	99.51%
Rejection rate for CIBIL < 550	89.64%

Interpretation

• The dataset has more approved than rejected applications, with an overall approval rate of approximately 62.22%.

• Average CIBIL score is materially higher among approved applications than rejected applications.

• Within this dataset, the CIBIL >700 group has a very high observed approval rate, while the CIBIL <550 group has a high observed rejection rate.

• Income, loan amount, and asset variables add context to borrower profiles but require appropriate data-quality checks.

11. Business Value

• Portfolio visibility through a clear summary of application outcomes.

• Risk exploration through comparisons across CIBIL score segments.

• Financial profile analysis using income, loan, and asset-based measures.

• Decision support through descriptive evidence presented in interactive reporting.

12. Conclusion

CredIntel demonstrates an end-to-end descriptive analytics workflow for a loan approval dataset. It combines data quality assessment, exploratory analysis, financial feature engineering, MySQL querying, and Power BI reporting to turn application-level records into interpretable lending insights.

The clearest observed separation in this dataset is between the CIBIL score profiles of approved and rejected applications. Findings are descriptive and dataset-specific; they do not establish causation, predict future decisions, or define lending policy.

13. Skills Demonstrated

Python • Pandas • NumPy • Data Cleaning • Exploratory Data Analysis • Financial Ratio Analysis • Feature Engineering • MySQL • SQL Aggregations • Data Visualization • Power BI • Dashboard Development • Business Insight Communication

14. Author

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From Loan Data to Credit Intelligence.